

Speech Emotion Recognition Using Wav2Vec2 and Linear SVM

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Abstract

Speech Emotion Recognition (SER) asks whether a machine can identify emotional state from speech signals even when the spoken words alone are not enough. In this project, we study SER using a lightweight but effective pipeline built around pretrained wav2vec 2.0 representations and a linear Support Vector Machine classifier. The design is intentionally simple. Instead of training a large end-to-end network, we freeze the speech encoder and study how much emotional information can be extracted through different temporal pooling strategies. We evaluate the system on two acted-speech datasets, RAVDESS and CREMA-D, under both in-domain and cross-domain conditions. The strongest in-domain result is obtained with mean pooling on RAVDESS, reaching 66.48 % accuracy, while cross-dataset testing drops to 33.81 %. The results show that pretrained self-supervised features are useful for SER, but cross-corpus generalization remains difficult.

Keywords

speech emotion recognition, wav2vec 2.0, LinearSVC, cross-domain generalization, pooling strategies

1 Introduction

Emotion in speech is carried by more than the literal meaning of words. A short utterance may sound angry, hesitant, sad, cheerful, or neutral depending on pitch, energy, rhythm, speaking rate, and voice quality. This makes Speech Emotion Recognition useful in applications such as conversational agents, call-center analysis, affective computing, assistive systems, and mental-health monitoring. In all these settings, the emotional tone of speech can provide information that is not available from text alone.

At the same time, SER is a difficult problem because emotion is subjective, speaker-dependent, and strongly affected by recording conditions. A model that performs well on one dataset may fail when tested on another dataset with different speakers, microphones, acting styles, or class distributions. This is especially important for acted emotional corpora, where the same label can sound different across datasets.

This project is motivated by that issue. We want a system that is strong enough to use modern speech representation learning, but simple enough to analyze clearly. Therefore, we use pretrained wav2vec 2.0 as a frozen feature extractor and train a linear SVM classifier on top of utterance-level pooled features. This setup allows us to study the role of the representation, pooling strategy, and dataset shift without hiding the behavior inside a large neural network.

The main contributions of this work are:

- We build a complete SER pipeline using frozen wav2vec 2.0 embeddings and LinearSVC.

- We evaluate the model on RAVDESS and CREMA-D using speaker-independent splits.
- We compare in-domain and cross-domain performance to measure generalization.
- We analyze four pooling strategies: Mean, Max, Mean+Std, and Mean+Std+Max.
- We discuss practical observations such as class imbalance, scaling, and domain shift.

2 Related Work

Traditional SER systems were mostly based on handcrafted acoustic features such as MFCCs, pitch statistics, energy, zero-crossing rate, spectral centroid, and other prosodic descriptors [1, 3]. These features are easy to compute and often interpretable, but they require careful manual design. Their performance can also degrade when the recording environment or speaker population changes.

Deep learning methods improved SER by learning features directly from spectrograms or waveforms. Convolutional and recurrent models can capture local spectral patterns and temporal structure more flexibly than handcrafted pipelines [4]. However, training deep models from scratch usually requires large labeled datasets, which are not always available in emotion recognition.

Self-supervised speech models such as wav2vec 2.0 and HuBERT changed this situation by learning strong speech representations from large amounts of unlabeled audio [2, 5]. These models produce frame-level embeddings that contain phonetic, acoustic, and speaker-related information. Since emotion is also reflected in acoustic patterns, these pretrained embeddings are useful for SER even without task-specific fine-tuning.

However, cross-corpus SER remains challenging. Different datasets often use different recording setups, actor populations, label definitions, and emotional intensities. As a result, a model trained on one corpus may not generalize well to another corpus [8]. Our project directly studies this issue by training on CREMA-D and testing on RAVDESS.

3 Methodology

3.1 Datasets and Label Space

We use two standard emotional speech datasets: RAVDESS and CREMA-D. RAVDESS contains acted emotional speech from 24 professional actors and is balanced across several emotion categories [6]. CREMA-D is larger, with 7,442 clips from 91 actors, and contains more speaker variation [7].

To make the two datasets comparable, we map both corpora into a common six-class label space:

{angry, disgust, fear, happy, neutral, sad}.

In RAVDESS, the classes calm and surprise are removed because they are not part of the shared label set.

Table 1: Datasets after mapping both corpora into the same six-emotion label space.

Dataset	Speakers	Clips	Train/Test	Labels
RAVDESS	24	1056	20 / 4 spk.	6
CREMA-D	91	7442	73 / 18 spk.	6

All experiments use speaker-independent splits. This is important because random clip splitting can place the same speaker in both training and testing sets, which may overestimate performance. In RAVDESS, speakers 1–20 are used for training and speakers 21–24 are used for testing. In CREMA-D, 73 speakers are used for training and 18 speakers are used for testing.

3.2 Preprocessing

Each audio file is converted to mono, resampled to 16 kHz, and peak-normalized. If x is the waveform, the normalized signal is

$$\tilde{x} = \frac{x}{\max(|x|) + \varepsilon}, \quad (1)$$

where ε is a small constant used for numerical stability. This step prevents very loud or very quiet recordings from dominating the model input.

The preprocessing is deliberately simple. We do not apply heavy denoising, augmentation, or silence trimming. This keeps the pipeline reproducible and makes the results easier to interpret.

3.3 Feature Extraction

After preprocessing, each waveform is passed through the frozen facebook/wav2vec2-base encoder. For an utterance with T frames, wav2vec 2.0 outputs

$$X = [x_1, x_2, \dots, x_T] \in \mathbb{R}^{T \times 768}. \quad (2)$$

Here, each frame vector x_t has 768 dimensions. The encoder is not fine-tuned on emotion labels. This means that all learning happens only in the downstream classifier.

Freezing the encoder has two advantages. First, it reduces computational cost. Second, it makes the experiment cleaner because performance differences mainly come from pooling and classification choices rather than unstable neural-network training.

3.4 Pooling Strategies

Since wav2vec 2.0 outputs a variable-length sequence, we need to convert it into a fixed-length utterance vector. We compare four pooling strategies.

Mean pooling is defined as

$$\mu = \frac{1}{T} \sum_{t=1}^T x_t. \quad (3)$$

Max pooling is defined component-wise as

$$m_j = \max_{t \in \{1, \dots, T\}} x_{t,j}. \quad (4)$$

Table 2: Core implementation choices used throughout the project.

Component	Setting
Audio	mono, 16 kHz, peak normalization
Encoder	frozen facebook/wav2vec2-base
Pooling	Mean / Max / Mean+Std / Mean+Std+Max
Scaling	StandardScaler on train split only
Classifier	LinearSVC, balanced class weights
Search	3-fold grid search over C values
Metrics	accuracy and macro-F1

We also test Mean+Std and Mean+Std+Max. The standard deviation vector is

$$\sigma = \sqrt{\frac{1}{T} \sum_{t=1}^T (x_t - \mu)^2}. \quad (5)$$

The feature dimensions are 768 for Mean, 768 for Max, 1536 for Mean+Std, and 2304 for Mean+Std+Max.

3.5 Classification

The classifier is LinearSVC with balanced class weights. For class c , the class weight is

$$w_c = \frac{N}{K n_c}, \quad (6)$$

where N is the total number of training samples, K is the number of classes, and n_c is the number of training samples in class c .

Before classification, all features are standardized using StandardScaler. The scaler is fitted only on the training split and then applied to the test split. This is important because fitting the scaler on test data would leak information.

The regularization parameter C is selected using 3-fold grid search over

$$C \in \{0.01, 0.1, 1, 10, 100\}.$$

4 Experimental Design

The experiments are divided into three main settings.

Task 1: RAVDESS in-domain evaluation. The model is trained on RAVDESS training speakers and tested on held-out RAVDESS speakers.

Task 2a: CREMA-D in-domain evaluation. The model is trained and tested on separate CREMA-D speakers.

Task 2b: Cross-domain evaluation. The model is trained on CREMA-D and tested on RAVDESS. This is the hardest setting because both speaker identity and dataset characteristics change.

Task 3: Pooling comparison. The same RAVDESS split is used to compare different pooling strategies.

We report accuracy and macro-F1. Accuracy measures the overall fraction of correct predictions, while macro-F1 gives equal importance to every class. Macro-F1 is especially useful here because neutral has fewer samples than other classes after label harmonization.

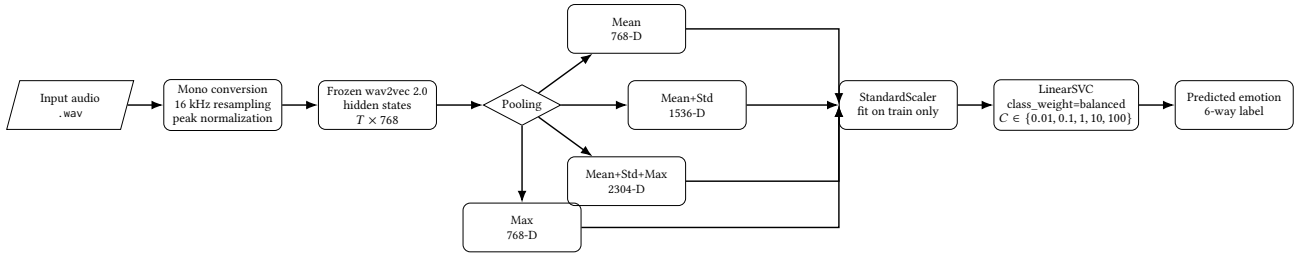


Figure 1: Project pipeline. The architecture uses frozen wav2vec 2.0 features, explicit pooling, standard scaling, and a linear classifier.

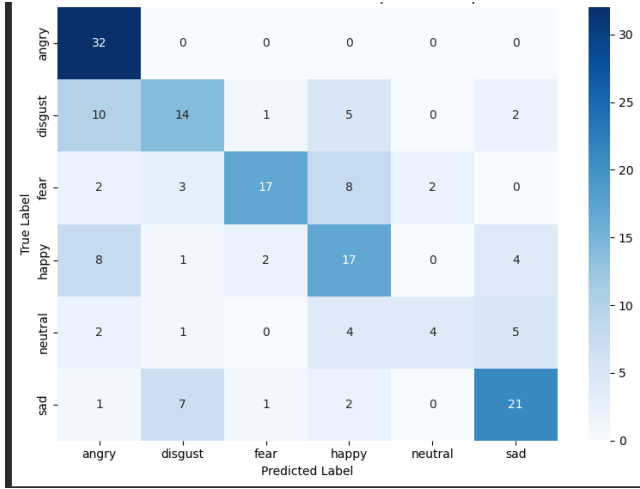


Figure 2: Confusion matrix for Task 1 on RAVDESS using the speaker-independent split. The model detects angry samples very strongly, while neutral remains the most difficult class.

Table 3: Main evaluation results across tasks.

Setting	Accuracy	Macro-F1
RAVDESS → RAVDESS	59.66%	0.5633
CREMA-D → CREMA-D	58.31%	0.5710
CREMA-D → RAVDESS	33.81%	0.3342

5 Results and Analysis

The in-domain results on RAVDESS and CREMA-D are similar. RAVDESS achieves 59.66 % accuracy and CREMA-D achieves 58.31 %. This suggests that the pipeline is stable across two different datasets when training and testing are done within the same corpus.

However, cross-domain performance drops sharply. When the model is trained on CREMA-D and tested on RAVDESS, accuracy becomes only 33.81 %. This shows that the model does not fully learn dataset-independent emotion cues. Instead, it also learns corpus-specific patterns.

The RAVDESS per-class results show that angry has the highest recall. This means the model detects all angry samples correctly in the test split. However, its precision is lower, meaning some non-angry samples are also predicted as angry. This agrees with

Table 4: Per-class results for the speaker-independent RAVDESS evaluation.

Emotion	Prec.	Rec.	F1	N
angry	0.58	1.00	0.74	32
disgust	0.54	0.44	0.48	32
fear	0.81	0.53	0.64	32
happy	0.47	0.53	0.50	32
neutral	0.67	0.25	0.36	16
sad	0.66	0.66	0.66	32
Macro	0.62	0.57	0.56	176

Table 5: Per-class cross-domain performance for CREMA-D → RAVDESS.

Emotion	Prec.	Rec.	F1	N
angry	0.35	0.30	0.32	192
disgust	0.46	0.38	0.42	192
fear	0.39	0.38	0.39	192
happy	0.28	0.40	0.33	192
neutral	0.25	0.47	0.32	96
sad	0.34	0.17	0.23	192
Macro	0.35	0.35	0.33	1056

Figure 2, where several disgust and happy samples are confused as angry.

Neutral is the weakest class, with only 0.25 recall. This is expected because neutral speech is subtle and has fewer samples. Fear has high precision but lower recall. This means that when the model predicts fear, it is often correct, but it misses many actual fear samples. Sad has balanced precision and recall, making it one of the more stable classes.

The cross-domain results are weaker for almost every class. Sadness drops the most, with an F1 score of only 0.23. Neutral has higher recall but low precision, which means the model predicts neutral too often. This kind of behavior usually happens when the model is uncertain and uses one class as a fallback.

The pooling study gives an important observation. Mean pooling achieves the best accuracy at 66.48 %. Mean+Std+Max reaches 63.64 %, Mean+Std reaches 59.66 %, and Max pooling performs worst at 50.00 %.

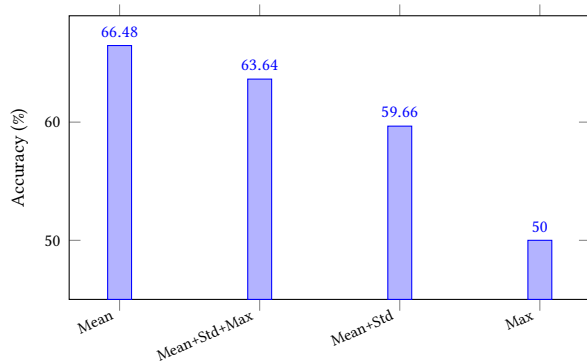


Figure 3: Pooling study on RAVDESS. Mean pooling gives the best result under a linear classifier.

This shows that adding more statistics does not always improve performance. A larger feature vector may contain more information, but it also increases dimensionality. Since the classifier is linear and the training data is limited, extra statistics may add noise or make the decision boundary less stable.

Mean pooling works well because it summarizes the overall emotional pattern of the utterance. Max pooling may focus too much on short local peaks, which may not represent the full emotional content.

6 Practical Observations

One practical observation is that feature scaling is important. During cross-domain testing, using a mismatched scaler reduced accuracy from 33.81 % to 26.99 %. This shows that normalization must be handled carefully, especially when training and testing datasets are different.

Another observation is that class imbalance affects performance. Even though balanced class weights are used, neutral remains difficult. This suggests that class weighting alone cannot fully solve subtle-class detection problems.

The results also show that speaker-independent evaluation is harder but more realistic than random splitting. If the same speaker appears in both train and test sets, the model may learn speaker-specific cues rather than emotion-specific cues.

7 Limitations

This project has some limitations. First, the wav2vec encoder is frozen. Fine-tuning may improve accuracy, but it would also require more computation and careful regularization. Second, the datasets are acted emotional corpora, which may not fully represent natural emotional speech. Third, the classifier is linear, so it may not capture complex nonlinear relationships in the embeddings.

Another limitation is that the cross-domain experiment uses CREMA-D to RAVDESS only. A more complete study would also test RAVDESS to CREMA-D and multi-corpus training. In addition, the current system does not use augmentation, attention, or domain adaptation, so there is still scope for improvement.

8 Future Work

There are several possible extensions. First, the system can be improved by fine-tuning wav2vec 2.0 on emotional speech. Second, attention-based pooling can be used instead of simple statistical pooling. Attention pooling may learn which frames are most emotionally important.

Third, domain adaptation can be used to reduce the gap between CREMA-D and RAVDESS. This is important because real-world speech emotion systems must work across different speakers and recording conditions.

Fourth, the model can be trained on a mixture of multiple datasets. Multi-corpus training may help the classifier learn more general emotional cues. Finally, the trained scaler, pooling method, and classifier can be exported as a single inference pipeline for predicting emotion from new .wav files.

9 Conclusion

This project demonstrates that a simple SER pipeline can perform reasonably well when built on top of modern pretrained speech representations. Frozen wav2vec 2.0 embeddings combined with mean pooling and a linear SVM give strong in-domain performance, especially on RAVDESS. However, the cross-domain results show that SER is still sensitive to dataset shift.

The main lesson is that representation quality matters, but it does not completely solve generalization. Mean pooling is surprisingly effective, while larger feature summaries do not always help. The final system is simple, reproducible, and interpretable, making it suitable for a student project and a useful baseline for future improvements.

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